

ADITYA PATIL

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SUMMARY

Product security engineer with 4+ years securing web, API, IoT, mobile, and LLM-integrated systems through source code review, reverse engineering, SAST/DAST at scale, threat modeling, and exploit development. Carries assessments from initial access through executive reporting and works with engineering teams to close findings, translating exploitability and business impact for technical and non-technical audiences. M.S. Cybersecurity candidate (GPA 3.92); OSCP exam scheduled Sep. 2026.

EXPERIENCE

Vulnerability Management Co-op – Rochester Regional Health **Aug. 2025 – July 2026**

- Configured, tuned, and triaged SAST and DAST tooling across 15+ production and test environments, partnering with application and infrastructure teams to drive 100+ security findings to closure.
- Planned and executed penetration tests and application security reviews on 6+ internal and external web and API applications in a HIPAA-regulated environment, delivering remediation guidance aligned to HIPAA and OWASP to the owning engineering teams.
- Applied threat modeling and security architecture review to new and changing services before deployment, prioritizing risk early and evaluating third-party vendors ahead of onboarding.
- Monitored the external attack surface of 5 subsidiary business units, responding to newly exposed assets and escalating recurring misconfigurations to owning teams through full resolution.
- Contributed to internal security standards and secure configuration guidance, advising application teams on policy compliance and handling of confidential data.

Security Researcher – Rochester Institute of Technology, SAFE Lab **Jan. 2025 – Present**

Product Security Assessment – Source Code Review, Reverse Engineering & Exploit Development (industrial IoT device)

- Assessed a Windows 10 IoT industrial sensing device and its .NET application stack, carrying the work from initial access through exploitation to executive reporting.
- Source code review of the decompiled .NET monitoring application (dnSpy, dotPeek) surfaced a hardcoded Azure Storage tenant key protected only by a custom encryption routine; reverse engineered that scheme to recover the key and demonstrated unauthorized access to the client's cloud storage tenant, user accounts, and all synced device telemetry.
- Identified a forgeable authentication design in which non-expiring tokens were derived from non-secret local values (hostname, username, hardware ID), enabling impersonation across the device fleet, a design-level flaw automated scanning does not reach.
- Discovered a kiosk-mode misconfiguration and weaponized it into a BadUSB payload that automated the escape sequence, gaining OS-level access to the filesystem and application binaries.
- Authored and presented a report covering 5 major security gaps, translating exploit chains into business risk and coordinating remediation guidance for both engineering and executive audiences.

External Attack Surface Assessments – Four Government Entities (three continents) & a Major HR/Payroll Provider

- Scoped and executed open-ended external attack surface assessments across five organizations, building asset inventories from zero via subdomain/DNS intelligence, ASN mapping, Internet Archive CDX history, Shodan, fingerprinting, and personnel OSINT.
- Correlated technology fingerprints with archived URLs to reach likely-sensitive endpoints, uncovering unauthenticated configuration files, exposed directories, and information disclosure.
- Reverse engineered 7 shipped iOS application binaries with Ghidra, identifying exposed private key material, TLS pinning weaknesses, and information disclosure in production code.
- Delivered 46 findings and 9 major security gaps, including payroll and PII exposure risk, through written reporting and recorded video walkthroughs for executive stakeholders.

Systems Engineer (Cybersecurity) – Tata Consultancy Services

June 2023 – June 2024

- Performed penetration testing on LLM-integrated applications, identifying OWASP Top 10 for LLM risks including prompt injection, sensitive information disclosure, excessive agency, and data/model poisoning.
- Built custom Python tooling to automate reconnaissance and finding validation across engagements, saving ~5 hours per penetration test.
- Identified and reported vulnerabilities across 25+ customer-facing web and API assets for a Big Four client, including XSS, CSRF, SQL injection, privilege escalation, and sensitive data disclosure.
- Conducted manual penetration testing with Burp Suite, covering authentication bypass, IDOR, and injection, to validate and prioritize real risk beyond automated scan output.
- Chained lower-severity vulnerabilities such as SQL injection into remote code execution, demonstrating real-world exploitability and business impact beyond individual finding severity.

Security Consultant Intern – SecureLayer7

March 2023 – May 2023

- Performed web and API penetration testing for telecom, information management, and healthcare clients, discovering critical XSS, CSRF, SQLi, and privilege escalation issues with OWASP/PTES-aligned remediation guidance.
- Discovered and responsibly disclosed vulnerabilities in open-source projects including Apache Zeppelin, coordinating with maintainers through remediation.

SKILLS & TOOLS

Languages: Python, JavaScript, C, C++, PowerShell

Application & Product Security: Source code review, reverse engineering, threat modeling, secure design & architecture review, exploit development, SAST/DAST, OWASP, PTES

AI & LLM Security: LLM and agentic application testing, prompt injection, OWASP Top 10 for LLM, MCP & tool-calling security, data/model poisoning

Security Tooling: Burp Suite, Caido, IBM AppScan, Rapid7 InsightVM / InsightAppSec, Ghidra, dnSpy, dotPeek, Metasploit, Nmap, Wireshark, Postman

Cloud, Infra & Recon: Azure Storage & tenant security, DNS/ASN mapping, Internet Archive CDX, Shodan, Wappalyzer, OSINT

Vulnerability Management: Scan configuration & tuning, triage, remediation tracking, third-party risk, secure configuration guidance

Coursework: Cryptography, Cyber Analytics & Machine Learning, Adv. Malware & Computer Forensics

AWARDS & ACHIEVEMENTS

- 2nd Place - Collegiate Penetration Testing Competition (CPTC) Great Lakes Regional 2025.
- Finalist - Collegiate Penetration Testing Competition (CPTC) Global Finals 2026.
- Top 100 out of 12000 candidates - TCS HackQuest Season 7 CTF Competition.
- Hall of Fame - Drexel University & State of California - reported critical vulnerabilities exposing PII/PHI

EDUCATION

Rochester Institute of Technology | M.S. Cybersecurity

Fall 2024 - Expected Dec. 2026 • GPA: 3.92/4.00

Savitribai Phule Pune University | B. E. Computer Engineering

June 2018 - April 2022 • GPA: 8.95/10.00

CERTIFICATIONS

OSCP - Offensive Security Certified Professional | In progress, exam scheduled Sep. 2026

PROJECTS

Linux Package Integrity with PQC Hybrid Signatures: verification system combining RSA-2048 with post-quantum ML-DSA-65.

Malicious URL Detection Using Machine Learning: logistic zregression model classifying URLs on lexical features.